

Anran Li

Add: Jiangsu Aidea Pharmaceutical Group Co., Ltd, Yangzhou, Jiangsu 225000, China Tel: +86 187 9588 3803

Email: anranli@stu.cpu.edu.cn Website: <https://lianran1998.github.io/>

EDUCATION

China Pharmaceutical University (CPU) Sep. 2020 – Jun. 2023

M.Sc. in New Dosage Forms and Technologies, GPA: 90.31/100.00

Shandong University of Traditional Chinese Medicine (SDUTCM) Sep. 2016 – Jun. 2020

B.Sc. in Pharmaceutical Engineering, GPA: 82.29/100 (Ranked 3/247)

PUBLICATIONS

Papers:

- A Li, K. Zhang, Z. Wang, S. Liu, X. Li, J. Ren, J. Zhang, Y. Shi, Y. Gao, *Hybrid Deposition Modeling Technology for Preparing Rocket-separation 3D-print Tablet with Greater Compliance for the Treatment of Helicobacter Pylori Infection*, Adv. Mater. Technol. 2025, 10, 2400840.
- K. Zhang, D. Zheng, A. Li, S. Liu, Z. Wang, Chen Yu, J Yu, Y Gao, J Zhang, *Recent Advances of Pharmaceutical 3D Printing Extrusion Molding Technology*, Acta Pharm. Sin. 58 (2023) 2640-2655.
- F. Ding, W. Cao, R. Wang, N. Wang, A. Li, Y. Wei, S. Qian, J. Zhang, Y. Gao, Z. Pang, *Mechanistic Study on Transformation of Coamorphous Baicalein-Nicotinamide to Its Cocrystal Form*, J. Pharm. Sci. 112 (2023) 513–524.

Patent:

- Y. Chen, A. Li, T. Hu, H. Fu. "A Pharmaceutical Formulation, and Its Preparation Method and Application. " CN 118384165 B, 2026. (Granted)
- Y. Gao, A. Li, K. Zhang, Z. Wang, J. Zhang, S. Qian, Y. Wei. "3D Printed Combined Tablet for Treating Helicobacter Pylori Infection. " CN 116019776 B, 2026. (Granted)
- J. Zhang, S. Liu, A. Li, F. Lv, F. Yuan, Y. Gao, S. Qian. "A 3D-Printed Olaparib Tablet and Its Preparation Method. " CN 115998699 B, 2025. (Granted)

Monograph:

- A. Li, H. Zhu, S. Qian. *Research on Formulated Granules of Traditional Chinese Medicine* [J]. China Yearbook of Traditional Chinese Medicine (Academic Volume), 2021, 16 (05): 423-424.
- M. Huang, A. Li, S. Qian. *Study on the Transdermal Properties of Chinese Medicine Plasters* [J]. China Yearbook of Traditional Chinese Medicine (Academic Volume), 2022, 17 (06): 425-426

RESEARCH

Research Area: 3D Printing Technology in Pharmaceutical Applications

Keywords: 3D Printing, Drug Delivery, HDM Printer, Oral Solid Dosage, Pharmaceutical Innovation

(1) 3D Printing Preparation Platform Design and Construction (Project Leader) Jul. 2021 – Jul. 2023

Supervisor: Prof. Jianjun Zhang

- Explored various 3D printing technologies and operation principles.
- Designed and assembled four structural 3D printers, including I3 and Core-XY models.
- Independently handled programming, debugging, firmware rewriting, and setup of the first 3D printing research platform in the laboratory.

(2) 3D Printed Combination Tablet for Quad-Therapy (Project Leader) Oct. 2021 – Jun. 2023

Supervisor: Prof. Yuan Gao

- Designed and built a customized HDM 3D printer to fabricate combination tablets for *H. pylori* quad-therapy, addressing poor patient compliance.
- Developed 'rocket-separation' combination tablets, achieving temporal-spatial controlled release for multi-drug delivery.
- Established HPLC assay and discriminatory dissolution methods to evaluate tablet CQAs and release kinetics.
- Published a peer-reviewed paper in Adv. Mater. Technol. and completed a master's thesis based on research findings.

(3) Development of 3D Printed Oro dispersible Tablet Feb. 2022 – Jun. 2023

Supervisor: Prof. Jianjun Zhang

- Employed advanced polymers and plasticizers to innovate beyond the original formulation, integrating hot-melt extrusion with 3D printing for enhanced drug delivery.
- Engineered and optimized the structure of orodispersible tablets, reducing their size to alleviate patient swallowing issues while maintaining therapeutic efficacy.

Research Area: Drug Formulation and Material Science

Keywords: Drug Formulation, Pharmaceutical Excipient, Generic Drugs

(1) Study on the Properties of Microcrystalline Cellulose Powder

Jul. 2021 – Dec. 2021

Supervisor: Prof. Jianjun Zhang

- Conducted a comparative study on the effects of different sources of microcrystalline cellulose on powder flowability, tablet compressibility and dissolution.

(2) Formulation Development of Generic Anti-cancer Drug

Sep. 2020 – Jun. 2021

Supervisor: Prof. Jianjun Zhang

- Engaged in the R&D pilot scale-up and methodological validation of quality standards for generic anti-cancer drugs.
- Completed stability studies comparing in-house samples with reference formulations.

WORK EXPERIENCE

Jiangsu Aidea Pharmaceutical Co., Ltd.

Position: Senior Formulation Scientist

Jul. 2023 – Present

Research Area: Formulation development of drugs for the treatment of HIV

(1) Development of Generic Antiviral 201 Tablet

- Executed pilot scale-up and stability testing, successfully ensuring product consistency, batch reproducibility, and critical quality attribute (CQA) alignment.
- Authored CTD Module 3 regulatory dossiers for drug registration and conducted rigorous data integrity/authenticity audits to guarantee compliance.
- Achieved successful Bioequivalence (BE) study completion against the reference listed drug (RLD), subsequently securing drug approval and commercial launch.

(2) Study on the Original Antiviral Drug 017 Tablet

- Conducted pre-formulation studies, evaluating API particle size distribution, hygroscopicity, and solid-state polymorphs..
- Investigating the solubilization capabilities of different types of lactose on insoluble drugs and selecting optimal processes for dosage form development, realized the ideal in vitro drug release effect and leading to patent publications.

(3) Development of Long-Acting Injectable & Oral Lead-In Systems

- Designed active formulations and matching placebos, optimizing solvent systems and excipient ratios.
- Executed formulation and process optimization for the oral lead-in tablets, successfully overcoming manufacturability challenges associated with high drug loading.
- Authored CTD Module 3 dossiers for IND filing, securing regulatory compliance and technical data integrity for clinical trial authorization.

PROFESSIONAL SKILLS

Instruments: HPLC, XRD, UV-Vis, FTIR, Laser Particle Sizing Tester, Dissolution Apparatus

Equipment: Skilled in assembling 3D printers, with the ability to operate pharmaceutical equipment such as granulators, fluidized beds, wet granulators, tablet presses, spray dryer, etc.

IT Skills: Proficient in Office, Zotero, Photoshop, Illustrator, Origin, Sketch UP, Simplify3D, ChemDraw

Language: Chinese Mandarin (Native), English (IELTS 5.5)

HONORS & AWARDS

- | | |
|--|-------------|
| • Bronze Award, The 2nd Global Digital Intelligence Education Innovation Competition (Peking University) | Nov. 2025 |
| • Aidea Pharmaceuticals Special Award: Outstanding R&D Organization | Jan. 2024 |
| • The Second Prize, CPU Scholarship | Oct. 2022 |
| • The First Prize, CPU Scholarship | Oct. 2021 |
| • Outstanding Research Poster, Jiangsu Graduate TCM Innovation Forum | Dec. 2020 |
| • Team First Prize, the Jiangsu Province Graduate Industrial Innovation Competition | Dec. 2020 |
| • The First Prize, CPU Freshman Graduate Student (Top 5%) | Sep. 2020 |
| • Shandong Province Outstanding Undergraduate Student Award (Top 1%) | Jun. 2020 |
| • SDUTCM First Prize Scholarship Winner for 3 Consecutive Years (Top 2%) | 2016 – 2019 |
| • SDUTCM Excellent Student Cadre Winner for 3 Consecutive Years | 2016 – 2019 |
| • Provincial Team Silver Award, “Internet+” Innovation and Entrepreneurship Competition | Sep. 2019 |
| • Outstanding Student, SDUTCM “Si Yuan Cup” Scholarships (Top 1%) | Sep. 2019 |
| • Gold Award, SDUTCM Outstanding Student Scholarship (Top 1%) | Dec. 2018 |
| • Provincial Team Third Prize, the 5th Shandong Tech Innovation Contest | Dec. 2018 |
| • Outstanding Work, Communist Youth League of China Social Practice Competition | Nov. 2017 |